Lecture 5: Diffusion Models and Stochastic Generative Dynamics

1. Beyond a Single Invertible Transformation

Normalizing flows generate molecular configurations using an invertible transformation,

$$\mathbf{x} = f_\theta(\mathbf{z}). \tag{1}$$

This provides both generation and explicit probability evaluation.

However, exact invertibility and tractable Jacobian determinants impose significant structural constraints.

Diffusion models take a different approach.

Instead of learning one transformation,

$$\mathbf{z} \to \mathbf{x}, \tag{2}$$

they construct a sequence of many small stochastic transformations,

$$\boxed{\mathbf{x}_0 \to \mathbf{x}_{t_1} \to \mathbf{x}_{t_2} \to \cdots \to \mathbf{x}_T.} \tag{3}$$

The forward process gradually destroys molecular structure until the distribution becomes simple, often approximately Gaussian.

A learned reverse process then reconstructs molecular structure from noise.

2. The Forward Diffusion Process

Let

$$p_0(\mathbf{x}) \tag{4}$$

denote the data distribution.

We define a stochastic process that progressively adds noise.

In continuous time, a general diffusion process may be written as

$$\boxed{d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{W}_t,} \tag{5}$$

where

- $\mathbf{f}(\mathbf{x}, t)$ is a drift term,
- $g(t)$ determines the noise strength,
- $d\mathbf{W}_t$ is an increment of a Wiener process.

A particularly simple example is pure diffusion,

$$d\mathbf{x} = g(t)\, d\mathbf{W}_t. \tag{6}$$

As time increases, fine structure in the distribution is progressively destroyed.

The goal is to choose the forward process so that, at sufficiently large T,

$$p_T(\mathbf{x}) \approx \mathcal{N}(0, \sigma_T^2 I), \tag{7}$$

or another simple reference distribution.

3. The Ink-Diffusion Analogy

Consider placing a drop of ink into water.

Initially, the ink concentration is localized.

Diffusion spreads the ink until the concentration becomes broad and nearly uniform.

The forward process is easy:

$$\text{localized structure} \to \text{diffuse distribution.} \tag{8}$$

Now imagine observing only the final diffuse state and asking:

> Can we construct a process that statistically reconstructs the original localized distribution?

This is analogous to the generative problem in diffusion models.

The important point is that we do not attempt to reverse each microscopic Brownian trajectory.

Instead, we construct a stochastic process whose probability distribution evolves backward in time.

4. From Stochastic Trajectories to Probability Evolution

The stochastic differential equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{W}_t \tag{9}$$

generates random trajectories.

However, the probability density

$$p_t(\mathbf{x}) \tag{10}$$

evolves deterministically according to the Fokker–Planck equation,

$$\boxed{\frac{\partial p_t}{\partial t} = -\nabla \cdot [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + \frac{1}{2} g(t)^2 \nabla^2 p_t(\mathbf{x}).} \tag{11}$$

The first term describes probability transport due to the deterministic drift.

The second term describes spreading due to diffusion.

Thus,

$$\boxed{\text{random microscopic trajectories} \quad \Longrightarrow \quad \text{deterministic evolution of probability density.}} \tag{12}$$

5. Why We Cannot Simply Reverse the Noise

For deterministic Hamiltonian dynamics, reversing all momenta can in principle reverse the trajectory.

A stochastic diffusion process is different.

The noise term

$$d\mathbf{W}_t \tag{13}$$

contains random information that is not recoverable by simply changing the sign of time.

Thus the reverse dynamics is not

$$dt \to -dt \tag{14}$$

applied naively to the forward stochastic equation.

Instead, the correct reverse process must be derived so that its probability density follows the time-reversed Fokker–Planck evolution.

6. Reverse-Time Stochastic Dynamics

For the forward process

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{W}_t, \tag{15}$$

the corresponding reverse-time stochastic differential equation has the form

$$\boxed{d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] dt + g(t)\, d\overline{\mathbf{W}}_t,} \tag{16}$$

where time is integrated from T toward 0.

$$\begin{aligned}
\frac{\partial p_t}{\partial t} &= -\nabla \cdot [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + \frac{1}{2} g(t)^2 \nabla^2 p_t(\mathbf{x}) && (17)\\
&= -\nabla \cdot [\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})] + g(t)^2 \nabla^2 p_t(\mathbf{x}) - \frac{1}{2} g(t)^2 \nabla^2 p_t(\mathbf{x}) && (18)\\
&= -\nabla \cdot \left[\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x}) - g(t)^2 \nabla p_t(\mathbf{x})\right] - \frac{1}{2} g(t)^2 \nabla^2 p_t(\mathbf{x}) && (19)\\
&= -\nabla \cdot \big([f(\mathbf{x}, t) - g(t)^2 \nabla \log p_t(\mathbf{x})] p_t(\mathbf{x})\big) - \frac{1}{2} g(t)^2 \nabla^2 p_t(\mathbf{x}). && (20)
\end{aligned}$$

The additional drift term is

$$-g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}). \tag{21}$$

The quantity

$$\boxed{\mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})} \tag{22}$$

is called the score function.

The score points in the direction in which the probability density increases most rapidly.

7. Physical Interpretation of the Score

Diffusion tends to spread probability from high-density regions toward low-density regions.

To reverse this spreading, the reverse dynamics requires a drift toward higher-probability regions.

The score

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \tag{23}$$

provides exactly this directional information.

Thus the additional drift can be interpreted as the correction required to reverse the probability flow generated by diffusion.

Conceptually,

$$\boxed{\begin{array}{l}\text{forward diffusion spreads probability,}\\ \text{the score directs reverse dynamics toward probable configurations.}\end{array}} \tag{24}$$

The reverse process does not reconstruct the exact forward microscopic trajectory.

It reconstructs the evolving probability distribution.

8. Connection Between the Score and Molecular Forces

The score has a particularly natural interpretation in statistical mechanics.

For a Boltzmann distribution,

$$p(\mathbf{x}) = \frac{1}{Z}e^{-\beta U(\mathbf{x})}. \tag{25}$$

Taking the logarithm,

$$\log p(\mathbf{x}) = -\beta U(\mathbf{x}) - \log Z. \tag{26}$$

Taking the gradient,

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\beta \nabla_{\mathbf{x}} U(\mathbf{x}). \tag{27}$$

Since the molecular force is

$$\mathbf{F}(\mathbf{x}) = -\nabla_{\mathbf{x}} U(\mathbf{x}), \tag{28}$$

we obtain

$$\boxed{\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \beta \mathbf{F}(\mathbf{x}).} \tag{29}$$

Thus,

$$\boxed{\text{score} = \beta \times \text{force}} \tag{30}$$

for an equilibrium Boltzmann distribution.

This provides a direct conceptual bridge between molecular simulation and diffusion models.

A score model can be viewed as learning a generalized force field associated with a probability distribution.

9. What Does the Diffusion Model Learn?

The forward process is specified by design.

Thus

$$\mathbf{f}(\mathbf{x}, t) \tag{31}$$

and

$$g(t) \tag{32}$$

are known.

The unknown quantity in the reverse-time dynamics is

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}). \tag{33}$$

A neural network is therefore trained to approximate the score,

$$\boxed{\mathbf{s}_\theta(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x}).} \tag{34}$$

Once the score is known, the reverse-time SDE can be simulated.

Sampling then proceeds as

$$\boxed{\mathbf{x}_T \sim p_T \quad \longrightarrow \quad \text{reverse diffusion} \quad \longrightarrow \quad \mathbf{x}_0 \sim p_{\text{data}}.} \tag{35}$$

10. Denoising Score Matching

The score

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \tag{36}$$

is generally unknown because the density itself is unknown.

However, the forward noise process is known.

For many common diffusion processes, we know the conditional distribution

$$p_t(\mathbf{x}_t|\mathbf{x}_0). \tag{37}$$

This enables training through denoising score matching.

The model is trained so that

$$\mathbf{s}_\theta(\mathbf{x}_t, t) \tag{38}$$

predicts the score associated with noisy samples.

A generic objective has the form

$$\mathcal{L}_{\text{DSM}} = \left\langle \left| \mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{x}_0) \right|^2 \right\rangle. \tag{39}$$

Thus training can proceed without knowing the normalized data probability density explicitly.

12. A Simple Gaussian Perturbation

Consider

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}, \tag{40}$$

where

$$\boldsymbol{\epsilon} \sim \mathcal{N}(0, I). \tag{41}$$

Then

$$p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_0, \sigma_t^2 I\right). \tag{42}$$

The conditional score is

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0}{\sigma_t^2}. \tag{43}$$

Since

$$\mathbf{x}_t - \mathbf{x}_0 = \sigma_t \boldsymbol{\epsilon}, \tag{44}$$

we have

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\boldsymbol{\epsilon}}{\sigma_t}. \tag{45}$$

Thus learning the score is closely related to learning how to remove the added noise.

This is why diffusion models are often described as denoising models.

13. Diffusion as Generative Sampling

Once the score model has been trained, generation begins from noise,

$$\mathbf{x}_T \sim \mathcal{N}(0, \sigma_T^2 I). \tag{46}$$

The reverse-time dynamics is then integrated,

$$d\mathbf{x} = \left[\mathbf{f} - g^2 \mathbf{s}_\theta(\mathbf{x}, t)\right] dt + g\, d\overline{\mathbf{W}}_t. \tag{47}$$

As t decreases,

$$p_T \to p_t \to p_0. \tag{48}$$

At the end,

$$\mathbf{x}_0 \sim p_{\text{data}}. \tag{49}$$

Thus,

$$\boxed{\text{noise} \to \text{reverse stochastic dynamics} \to \text{molecular configuration.}} \tag{50}$$

15. Normalizing Flows Versus Diffusion Models

Normalizing flows and diffusion models solve closely related problems.

Both transform a simple probability distribution into a complex data distribution.

However, they use different strategies.

	Normalizing flow	Diffusion model
Transformation	deterministic	stochastic
Number of steps	one composed invertible map	many diffusion steps
Invertibility	explicitly required	not required as one global map
Density evaluation	direct through Jacobian	less direct
Main learned object	coordinate transformation	score function
Sampling	typically fast	typically iterative

The common principle is

$$\boxed{\text{transport probability from a simple distribution to a complex one.}} \tag{51}$$

18. Generative Models Versus Molecular Dynamics

Molecular dynamics and generative models both produce molecular configurations, but their interpretation is different.

Molecular dynamics generates a physical or approximate dynamical trajectory,

$$\mathbf{x}(0) \to \mathbf{x}(t). \tag{52}$$

A generative model seeks primarily to reproduce a probability distribution,

$$p_\theta(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x}). \tag{53}$$

A generated sequence of configurations therefore does not generally represent physical time evolution.

This distinction is essential:

$$\boxed{\begin{aligned} \text{molecular dynamics} &: \text{learn or specify forces and propagate physical time,} \\ \text{generative modeling} &: \text{learn probability transport and generate samples.} \end{aligned}} \tag{54}$$